

Master Thesis — Progress Report

Spain's MTU15 reform: empirical assessment in progress

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Outline of the talk

- **1. Reminder.** Objective, research question, context.
- **2. What I have done.** Data, methods, headline findings.
- **3. State of the work.** Claim portfolio, theoretical model.
- **4. Difficulties encountered.** Identification, confounds, surprises.
- **5. What is left.** Empirical, theoretical, writing.

Format

18–20 min, following advisor's guidance: a progress report rather than a results talk.

Objective and research questions

Aim

Empirically assess the impact of Spain's 2024–2025 **MTU15 reform sequence** on market outcomes and firm strategic behaviour, and extend the theoretical framework to capture the mechanism.

Five concrete sub-questions:

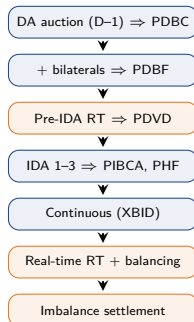
1. Does the **asymmetric-clock window** (DA60 + ID15 + ISP15) generate a measurable *dual-pricing redistribution* from BRPs to the TSO? Does it close at MTU15-DA?
2. Does finer market-time-unit granularity **reduce strategic forward-quantity withholding** by dominant firms, in line with the Allaz–Vila / Ito–Reguant prediction?
3. Does the reform create **heterogeneous winners and losers** across firms (Big-4 vs Fringe) and technologies (CCGT vs hydro vs nuclear)?
4. Does the per-segment incidence of imbalance settlement match a **Pigouvian benchmark**, or does the uniform dual-pricing rule misallocate the burden?
5. Has the **post-blackout operational regime** (*operación reforzada*) broken the separability of the reform sequence, and how do we identify each effect?

Big picture: the Spanish electricity market

For each delivery hour, the wholesale market clears **sequentially** across D-1 and D:

- **[OMIE]** Day-ahead auction (DA) $\Rightarrow$ PDBC
- + bilateral contracts $\Rightarrow$ PDBF
- **[REE]** pre-IDA technical restrictions $\Rightarrow$ PDVD
- **[OMIE]** Intraday auctions IDA-1, IDA-2 (D-1) and IDA-3 (D)
- **[OMIE]** Continuous market (XBID/SIDC), opens after IDA-2
- **[REE]** Real-time technical restrictions and balancing
- **Imbalance settlement:** ex-post, dual-pricing rule (EBGL Art. 52)

Each clearing produces a *firm program* that the next can adjust.



The 2024–2025 reform sequence

- Traditionally the market time unit (MTU) was **60 min** everywhere.
- The clock rewiring happened in **three stages**:
 - Dec 2024 — ISP15**: imbalance settlement becomes 15-min.
 - Mar 2025 — ID15**: intraday auctions and continuous become 15-min.
 - Oct 2025 — DA15**: day-ahead becomes 15-min.
- Two *transitional regimes* where settlement and trading granularities differ:
 - ISP15-win** (Dec 2024 – Mar 2025): DA60/ID60 + ISP15
 - DA60/ID15** (Mar 2025 – Oct 2025): DA60 + ID15 + ISP15

Period	ISP	DA	ID
Pre-Dec 2024	60	60	60
Dec24–Mar25	15	60	60
Mar25–Sep25	15	60	15
Oct25 onward	15	15	15

Minutes; 60 = hourly, 15 = quarter-hourly.

Plus an *operational regime overlay*: the **post-blackout *operación reforzada*** (Apr 28, 2025 onward) keeps tightening REE security criteria via PO-3.2 RRTT.

Data pipeline — scope

Four sources, strictly separated, all version-controlled and idempotent.

1. OMIE (Iberian market operator) — the primary source

25+ data families, 2018-01 → 2026-02 (8 years, 2 months) at native granularity (MTU60 pre-2025-03-19, MTU15 after):

- *Prices*: day-ahead (`marginalpdbc`), intraday auctions (`marginalpibc`), continuous market (`precios_pibcic`, `..._ronda`).
- *Programs*: `pdbc` (DA cleared), `pdbf` (+ bilaterals, 166M rows), `pdbce` (per firm), `pibca/pibci/pibcie` (intraday auctions), `pibcic/pibcac/pibcice` (continuous), `phf/phfc` (final hourly programs).
- *Bid curves and offer detail*: `cab/det` (DA offer headers + tranches), `icab/idet` (intraday), `orders/trades` (XBID limit orders + matched transactions), `curva_pbc/pibc` (aggregate supply-demand).
- *Capacities and annulments*: interconnection capacity (`capacidad_inter_*`), annulled offers/sessions.

2. REE / ESIOS (Spanish system operator)

Settlement detail (per-firm, per-segment), real-time RT prices (TipoRedespacho 61), balancing offers, unit outages, secondary-band offers, per-BSP aFRR (`liquicierre`, `liquicierresrs`).

Data pipeline — the rest

3. ENTSO-E Transparency Platform (pan-European)

Spanish control area (10YES-REE---0), API-based pull:

- *Prices and balancing*: A44 (DA prices, FR + ES), A85 (imbalance prices), A86 (imbalance volumes), A87 (FRR settlement, A01 TSO-paid, A02 BRP-paid).
- *Generation and load*: A75 (generation per type, hourly Spain), A73 (generation per unit), A65 (load actual), A69 (load forecast), A80 (wind+solar forecast), A72 (reservoir filling, weekly).
- Day-ahead forecasts of load and wind+solar, and intraday updates — canonical *forecast-error* controls.

4. AEMET (weather)

Temperature, wind, solar irradiance — fundamentals controls and instruments for residual-demand-slope identification.

Discipline

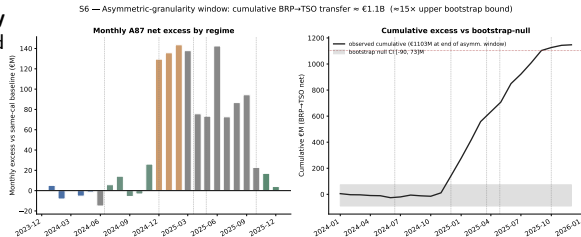
Raw → processed → derived layers strictly separated (raw never modified). One parser per family, all post-parse validation checks. Analytical outputs (regressions, figures) live outside data/. Symlinks to external SSD for the heavy raw layer.

Empirical strategy

- **No clean external counterfactual:** most comparable European markets adopted MTU15 already and differ structurally.
- Reduced-form before/after with rich controls:
 - Fundamentals: load, wind, solar (actual + forecast errors), temperature, fuel/CO₂.
 - Calendar structure: cal-month FE, day-of-week, hour-of-day, year FE (where appropriate).
 - Reform-date indicators (Dec 2024 / Mar 2025 / Oct 2025) and *reforzada* indicator.
- **Same-calendar-month robustness:** mandatory for any cross-regime claim (Spanish electricity is highly seasonal).
- **Cluster-robust SE:** by date or week-of-sample; Cameron–Miller $G \geq 30$ rule respected.
- **Cross-country placebo:** France DA prices flat across Spanish reform dates.
- **Structural-time-series counterfactual** (Brodersen et al. 2015) on key outcomes *ongoing*.

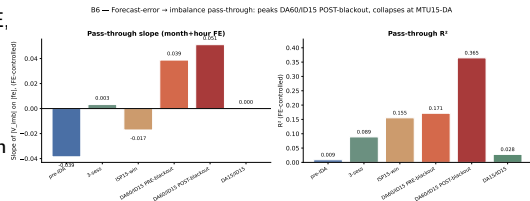
Headline finding 1 — system-layer settlement transfer

- During the **10-month asymmetric-granularity window** (Dec 2024–Sep 2025), the day-ahead market still cleared at 60-min while intraday and imbalance settlement had moved to 15-min.
- Net BRP→TSO settlement transfer: **+€1,094.9M** cumulative, vs same-calendar pre-reform baseline.
- Bootstrap null CI [-90, +73] M $\Rightarrow \approx 15\times$ the upper bound.
- **Collapses to €7.4M/mo** once the day-ahead aligns to 15-min (12 $\times$ drop) — the asymmetry was the source.
- This is a *regulatory redistribution* (BRPs $\rightarrow$ TSO $\rightarrow$ consumers via tariff), not a deadweight loss.



Headline finding 2 — the mechanism (forecast-error pass-through)

- Pooled regression: $|V_{imb,t}| \sim |fe_t| + R \times |fe_t| + FE$, daily grain, cluster SE by year-month.
- (V_{imb} = aggregate imbalance volume; fe = wind+solar forecast error; R = regime indicator.)
- Net per-regime slope β (GWh imbalance per GWh forecast error):
 - Pre-reform ~ 0
 - 3-session IDA (post 2024-06) ~ 0
 - ISP15-win ≈ -0.03
 - **DA60/ID15 $\approx +0.09$** — uniquely elevated
 - DA15/ID15 ~ 0
- **Robust to year FE.** The asymmetric-window elevation is the structural break.



Headline finding 3 — firm-level market power

- Synthetic-firm replication *à la* Ciarreta–Espinosa (2010).
- Substitute each Big-4 plant's offer with a same-technology, same-capacity competitive-fringe plant; re-clear the auction.
- **Iberdrola alone** accounts for $\sim \text{€}820\text{M}$ day-ahead cleared-price-difference rent post-MTU15-IDA, $\sim 98\%$ of the joint Big-4 transfer.
- Hydro 64% / CCGT 36%. Survives sensitivity tests (BRP migration, strict-hydro matching, blackout split, complex-offer exclusion).
- **Regime-invariant:** the rent does *not* arise from MTU15 reform; it is a persistent structural feature of the Spanish day-ahead market.

Behavioural complement:

- Big-4 voluntary intraday repositioning (q_2 , *à la* Ito–Reguant) shows a progressive collapse across regimes; partial recovery once the day-ahead aligns to 15-min.
- Robust to (i) day-ahead bilateral-channel controls, (ii) year FE, (iii) Fabra–Imelda-style spec with demand and wind/solar forecast-error controls + week-of-sample cluster: regime coefficients vary $< 0.1\%$ across specifications.

Tests of theoretical predictions — what worked

Ito & Reguant (2016, AER): sequential markets, market power, and arbitrage

Their two-period framework predicts that if firms have market power they over-supply forward ($q_1 > q_2$) and a finer spot grid attenuates the strategic value of forward commitment.

Confirmed. At the firm-day grain, Big-4 voluntary intraday repositioning (q_2) collapses progressively through the reform sequence; partial recovery once DA15 restores symmetric clocks. Robust to bilateral-channel controls, year FE, and Fabra–Imelda forecast-error controls (regime coefficients vary $< 0.1\%$). *Anchor for the behavioural side of the thesis.*

Ciarreta & Espinosa (2010, J Regul Econ): synthetic-firm method for Spanish market power

Substitute each Big-4 plant's offer with a same-tech, same-capacity competitive-fringe plant; re-clear the auction.

Replicated and extended. Iberdrola alone $\sim \text{€}820\text{M}$ day-ahead cleared-price-difference rent post-MTU15-IDA (98% of joint Big-4 transfer); hydro 64% / CCGT 36%. Survives BRP-migration, strict-hydro matching, blackout-split, and complex-offer caveats. Two-decade pattern persistence: Ciarreta–Espinosa documented Big-4 above synthetic-Big-4 in 2002–2005 at $\sim 21\%$; we measure $\sim 13\%$ post-MTU15-IDA (consistent with tighter regulation).

Tests of theoretical predictions — what did not work

Allaz & Vila (1993, JET): forward markets dispel oligopoly rents

Test: estimate $\beta = \partial \Delta Q_{IDA} / \partial q_{DA}$ at firm-period level; AV predicts $\beta < 0$ if firms commit forward to dispel future rents.

Rejected as a test. Within-unit, $Q_{actual} \approx q_{DA} + \Delta Q_{IDA}$ is a *mechanical accounting identity* $\Rightarrow \beta$ is mechanically near -1 minus the real-time-deviation share, regardless of strategic conduct. Not a structural slope. The thesis modelling chapter cannot lead with this; the surviving Allaz–Vila evidence comes from the Ito–Reguant q_2 trajectory above.

Hortaçsu & Puller (2008, RAND): sophisticated firms bid close to FOC-implied markup

Test: compare the implied Cournot–FOC Lerner against the realised marginal-bid markup.

Strategic-conduct interpretation rejected. For the leading firm at DA60/ID15, implied Lerner ≈ 0.93 vs realised marginal-bid Lerner ≈ 0.34 (ratio ≈ 0.36); cross-firm: realised markup is fairly uniform ($\approx \text{€}13\text{--}26/\text{MWh}$), implied is wildly heterogeneous (0.008–0.93). The implied measure is *formula-mechanical* (size, supply slope), not realised conduct. Regime coefficients survive but cannot be cited as strategic markup.

Bushnell (2003, JIE): hydro market power via reservoir shadow-price

Test: regress IB hydro top-quartile dispatch concentration on reservoir level.

Mechanism rejected. $\beta(\text{reservoir}) \approx 0$, $p \approx 0.87$; the placebo (Fringe firms) actually shows a stronger reservoir gradient than IB. The IB hydro Q4 concentration is real and persistent (+17pp gap vs Fringe, regime-invariant 2018–2025), but the underlying mechanism is *not* classical Bushnell shadow-pricing.

State of the empirical work

- I track findings formally (claim, status, evidence, robustness). Currently:
 - ~40 **alive** findings — pass all documented robustness checks.
 - ~7 **wounded** — survive in narrowed form; cited with caveat.
 - ~20 **dead** — retracted; kept as audit trail.
- Discipline: each cited number is regression-backed at proper grain (firm-day or finer), date- or week-clustered SE, and survives multi-spec robustness (sparse-FE baseline + augmented exogenous controls).

Three distinct policy levers identified:

1. **Clock-symmetry**: *implemented* at the final reform stage; closes the asymmetric-window transfer.
2. **Pigouvian incidence**: *open*. Wind + free-market retailers consistently bear 60–65% of imbalance settlement € regardless of clock symmetry.
3. **Operational regime overlay**: *operación reforzada* — the post-blackout reinforced-operation regime; a missing-market patch the wholesale layer cannot price. New zonal reactive-power market (Jun 2025) is the first market-design response.

Theoretical model — where it stands

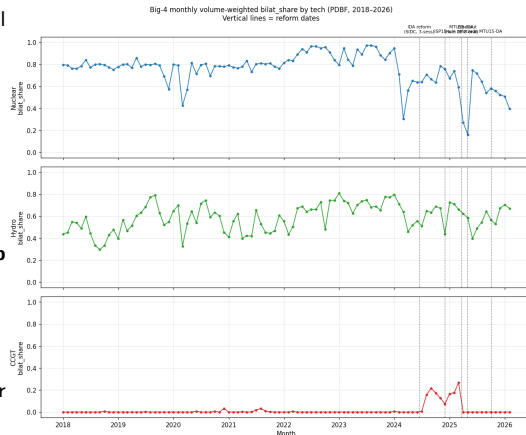
- Started from my Energy Economics research proposal: an extension of **Ito–Reguant (2016)** sequential-market model with an explicit role for the MTU mismatch.
- Predicted (i) DA15 is the key reform that smooths imbalances; (ii) dispersion risk concentrates in DA60/ID15; (iii) finer granularity creates winners and losers across heterogeneous portfolios.
- Empirically, all three predictions are **borne out qualitatively**; the magnitudes are larger than the theory suggested.
- **Pivot**: the project focuses on *very solid empirics* + a *simplified IO theory model* to illustrate the mechanism + a light structural exercise (a first-order-condition-based test, in the spirit of Hortaçsu–Puller and Allaz–Vila).
- **Not** a fully estimated structural auction model (Hortaçsu–Puller / Wolak): out of scope for the time available.

Difficulty 1 — identification under multiple confounds

- Each reform date has at least one competing event:
 - **Dec 2024** (imbalance-15-min) — concurrent with European gas-price normalisation.
 - **Mar 2025** (intraday-15-min) — concurrent with the CNMC's elimination of the bilateral-contract opportunity-cost buy-back obligation (a known rule change affecting day-ahead bidding).
 - **Apr 28, 2025** Iberian blackout $\Rightarrow$ *operación reforzada* (REE's reinforced-operation regime, continuous since).
 - **Oct 2025** (day-ahead-15-min) — partial overlap with reforzada.
- **Identification strategy:**
 - *Market-chain placement*: reforzada acts *after* the day-ahead clearing, so day-ahead-side outcomes face it only through bidder expectations; final-program outcomes face it mechanically.
 - *Reforzada-held-constant pairings*: imbalance-15-only $\rightarrow$ DA-60+ID-15 (both pre-blackout) for the intraday reform; DA-60+ID-15 (post-blackout) $\rightarrow$ full-MTU15 (both under reforzada) for the day-ahead reform.
 - Same-calendar-month robustness; cross-country placebo (France); bootstrap CIs.
- Honest record of **14 dead identification attempts**: two-way-FE DiD, wind-IV, profile-matching — each rejected after placebo / parallel-trends / randomization tests.

Difficulty 2 — a structural break we cannot yet explain

- Big-4 nuclear bilateral-contract share (out of total day-ahead sell volume) was stable at 80–95% throughout 2018–2023.
- It drops sharply in **Feb–Mar 2024**, three months *before* any MTU15 reform.
- Hydro shows the same Q1-2024 break (driven by auction-side growth); CCGT is unaffected (essentially never bilaterally committed).
- Concurrent with the **expiry of the Iberian gas-cap mechanism on 2023-12-31** and with Spain's first major renewable-saturation event (negative day-ahead prices emerge in April 2024).
- Regression of bilateral share on the negative-price-hours share (an exogenous proxy for renewable saturation), with unit/cal-month FE:
 $\beta = -3.05$ for nuclear ($p \approx 10^{-179}$, $R^2 = 0.13$);
the implied magnitude matches the observed drop.
For hydro: $\beta \approx 0$.



Mechanism unidentified — likely renewable-saturation onset interacting with multi-year PPA structures, but causal identification still open. This finding reframes how I read 2024 pre-reform data.

Difficulty 3 — *operación reforzada* as missing-market patch

- After the 2025-04-28 blackout, the system operator (REE) imposes a continuous **reinforced-operation regime**:
 - Forced redispatch of $\sim 20\text{--}30$ CCGTs + nuclear daily for security (technical-restrictions market).
 - Enlarged secondary reserve band; new zonal reactive-power market (Jun 2025).
 - Direct cost $\sim \text{€}666\text{M}$ (May 2025 – Mar 2026); technical-restrictions cost reached $\text{€}3.77\text{B}$ in 2025 (+49% YoY).
- Affects **all post-blackout regressions**, but in a structurally identifiable way:
 - Where the outcome is computed at the day-ahead clearing, *reforzada* enters only via bidder expectations.
 - Where it is computed at the post-RT final program, *reforzada* enters mechanically.
- **Cleanest empirical separator**: the asymmetric-window settlement transfer collapses at the day-ahead 15-min reform *while reforzada is still active* $\Rightarrow$ separates clock-symmetry (lever 1) from operational overlay (lever 3).

What is left — empirical

- **Q1 2024 mechanism:** pin down what specifically about the gas-cap expiry / renewable-saturation onset / PPA renegotiations drove the bilateral-channel collapse. Likely a self-contained chapter discussion.
- **Robustness re-pass** on the remaining alive findings (firm-level Lerner heterogeneity, bid complexification, post-blackout CCGT/aFRR market shares) using the same canonical conventions I have applied to the headline numbers.
- **Structural-time-series counterfactual** (Brodersen et al. 2015) on the headline outcomes: arbitrage wedge, intra-product dispersion, liquidity, balancing volumes.
- **Per-segment settlement verification** using the segment-level settlement file from REE (parsed but not yet used).
- **Figure polishing** for the thesis text.

What is left — theoretical and writing

- **Theoretical model:**

- Refine the Ito–Reguant extension to motivate the *three-lever* framing (asymmetric clock / Pigouvian incidence / operational overlay).
- Possibly add a small heterogeneous-firm extension to motivate the cross-technology bilateral-channel heterogeneity.
- Light **structural exercise**: Hortaçsu–Puller-style FOC test on Big-4 bid functions post-reform, without a full structural estimation.

- **Thesis writing:**

- Part I (system-layer reform impact) — drafting starts the week after this talk.
- Parts II–V (firm-level market power, cross-market specialisation, post-blackout strategic conduct, behavioural appendix) — content already drafted in the proposal.
- Deadline mid-June 2026.

- **Risk:** 8-week timeline is tight; the Part-II firm-level depth is the binding constraint.

Thank you.

Questions and feedback welcome.

Pablo Paramio -- CEMFI -- May 2026